Scheme of Valuation 25900560

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSEPHY202 - INTRODUCTION TO SPACEPHYSICS

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes Maximum Marks: 50

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions - Answer any 10 questions - Each question carries 2marks

1. Differentiate between aeronautics and astronautics with one example each.

At the altitude of 100 km from the mean sea level of the Earth, the atmosphere is thinner by several orders of magnitude as compared with the density of the atmosphere near the mean sea level that it becomes indistinguishable from the space vacuum. Hence the speed required for an aircraft to generate enough lift to maintain flight becomes impractical. Instead, any space-vehicle at that altitude or higher must rely on different propulsion systems to manoeuvre and maintain orbit. Aeronautics deals with flight below this level (FLIGHT OF AIRCRAFT) and astronautics with the flight above this altitude (FLIGHT OF ROCKET).

Correct answer – 2 Marks. For any other correct/ sensible answer/ explanation, give marks accordingly

2. What is the heliosphere?

The heliosphere is the vast region of space around the Sun that is influenced by the solar wind and the Sun's magnetic field. It acts like a protective bubble, surrounding the entire solar system and separating it from interstellar space

Correct answer – 2 Marks. For any other correct/ sensible answer/ explanation, give marks accordingly

3. Why space weather is monitored continuously?

The interest in space weather is primarily due to the rapid growth of the commercial satellite communications industry and the development of continental-sized power and communication grids. With these developments, we have become more susceptible to space weather storms through our reliance on hi-tech information systems and our growing global interconnections. Extreme space weather can damage critical infrastructure, which can lead to cascading failures in

other infrastructure systems including power, communication, water, healthcare, and transportation. With time, space weather will play a larger role in our everyday lives.
Correct answer – 2 Marks. For any other correct/ sensible answer/ explanation, give marks accordingly

4. What is limb darkening in photosphere?

The photosphere is the visible surface of the Sun. Since the Sun is a ball of gas, this is not a solid surface but is actually a layer about 600 km thick. When we look at the center of the disk of the Sun we look straight in and see somewhat hotter and brighter regions. When we look at the limb, or edge, of the solar disk we only see through the upper, cooler and dimmer regions. This explains the "limb darkening" that appears as a darkening of the solar disk near the limb.
Correct answer – 2 Marks. For any other correct/ sensible answer/ explanation, give marks accordingly

5. What is a geomagnetic storm?

A geomagnetic storm is a temporary disturbance of the Earth's magnetosphere caused by a solar wind shock wave and/or cloud of magnetic field that interacts with the Earth's magnetic field. Geomagnetic storm is defined by the intensification of the ring current and is one of the dominant modes of response of the magnetosphere to energy transfer from the solar wind.
Correct answer – 2 Marks. For any other correct/ sensible answer/ explanation, give marks accordingly

6. State the typical speed range of fast and slow solar wind.

The solar wind is a continuous flow of charged particles. It is supersonic in nature. The fast solar wind has flow speeds between 400 km/s and 800 km/s. The slow solar wind has lower speeds between 250 km/s and 400 km/s.
Correct answer – 2 Marks. For any other correct/ sensible answer/ explanation, give marks accordingly

7. What is meant by solar cycle?

The solar cycle is the periodic change in the number of sunspots and solar activity on the Sun's surface, caused by the variation in its magnetic field. It repeats approximately every 11 years, from one solar maximum to the next.
Correct answer – 2 Marks. For any other correct/ sensible answer/ explanation, give marks accordingly

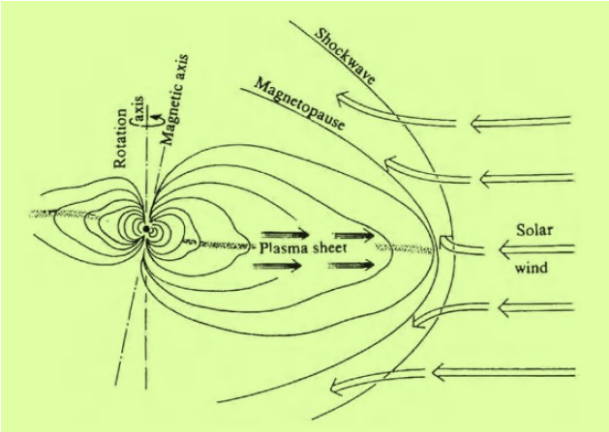
8. Explain one difference in orbit design between Aditya-L1 and Parker Solar Probe]

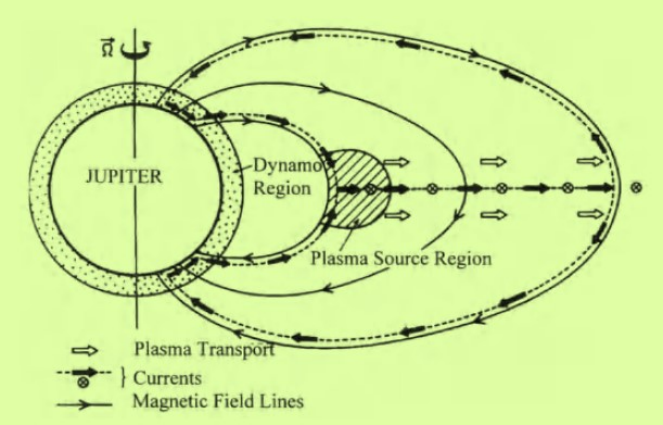
Aditya-L1 : Aditya-L1 Located at the Sun–Earth L1 Lagrange point, roughly 1.5 million km from Earth towards the Sun.
Parker Solar Probe: The spacecraft will travel directly into the Sun's atmosphere about 4 million miles (6437376 km or 7 million km) from our star's surface.
Correct answer – 2 Marks. For any other correct/ sensible answer/ explanation, give marks accordingly

9. Define an ionogram.

Ionogram is a graph showing the virtual height of the ionosphere versus the frequency of the radio waves. . It represents the time delay of the radio signal emitted by the ionosonde and reflected by the ionosphere, for a specific frequency.
Correct answer – 2 Marks. For any other correct/ sensible answer/ explanation, give marks accordingly

10. Draw a schematic of magnetosphere of Jupiter.





Any of these pictures – 2 Marks

11. What is scale height? Mention its significance.

The scale height (H) of the atmosphere is the distance over which the pressure or density falls by a factor of e.
By equation H = kT/mg , where k is Boltzmann's constant, T is the temperature, m is the molecular mass and g, the acceleration due to gravity.
H is one of the most useful parameters in the study of planetary atmospheres as it gives a convenient way of characterizing the atmosphere. It is a key parameter in understanding and modeling planetary atmosphere.
Correct answer – 2 Marks. For any other correct/ sensible answer/ explanation, give marks accordingly

12. What is meant by frozen in condition of IMF?

The frozen-in condition of the IMF refers to the concept that the interplanetary magnetic field (IMF) lines are "frozen into" the highly conductive solar wind plasma. In the case of solar wind, since the plasma has high energy density and the conductivity infinite, the field is frozen-in and carried away by the plasma. Hence the interplanetary magnetic field is frozen-into the solar wind.
Correct answer – 2 Marks. For any other correct/ sensible answer/ explanation, give marks accordingly

13. Explain recombination in the ionosphere.

Recombination in ionosphere is the process in which the free electrons combine with the positive ions, neutralizing them and forming a neutral atom or molecule. The main two different types are

Dissociative recombination of molecules and Radiative Recombination of Atomic Ions. In dissociative recombination molecular ions dissociate into possible excited states of the recombination products. In radiative recombination atomic ions directly recombine with free electrons and the excess energy is either entirely or partly expended as radiation (photon).
Correct answer – 2 Marks. For any other correct/ sensible answer/ explanation, give marks accordingly

14. Explain how TEC cause ionospheric delay?

Ionospheric delay is the time delay experienced by the satellite signals to travel through the Earth's ionosphere. The ionosphere acts as a dispersive medium causing velocity changes to the signals as it passes through it. Thus the variations in Total Electron Content (TEC) directly affect the delay of signals passing through the ionosphere: the higher the TEC, the larger the signal propagation delay (group delay).The relationship between the ionospheric delay (d_{ion}) and TEC is defined by the formula:

$$d_{ion} \propto \frac{TEC}{f^2}$$

where *f* is the signal frequency.

Correct answer – 2 Marks. For any other correct/ sensible answer/ explanation, give marks accordingly

[2x10=20]

Part B

Short Essay Answer Type Questions - Answer any 6 questions - Each question carries 5 marks

15. What are the main characteristics of plasma and how it differs from solids, liquids, and gases?

Plasma, often called the fourth state of matter, is critically significant in space physics because it constitutes the majority of matter in the universe. Unlike solids liquids and gases plasma consists of charged particles formed due to the stripping of electrons from the neutral atoms. Plasma contains almost equal amount of positive and negative charges, large enough to enable EM interaction. A plasma can interact with itself. Hence, plasma is not only controlled by external electric and magnetic fields, but also creates fields. These modify the external fields and the motion of the charged particles. Understanding plasma behavior is essential for comprehending space phenomena like solar flares, the solar wind, Earth's magnetosphere, and the aurora borealis. Studying plasma dynamics allows scientists to model and predict space weather, which can impact satellites, communication systems, and even power grids on Earth.
Correct answer – 5 Marks. For any other correct/ sensible answer/ explanation, give marks accordingly

16. Compare the contributions of the Space Race era with the collaborative era of space exploration.

The Space Race was a 20th-century rivalry between the United States and the Soviet Union to achieve technological and ideological superiority through spaceflight. The Space Race was a key aspect of the Cold War between the U.S. and the Soviet Union. The competition to develop

advanced rocket technology began shortly after World War II, as both superpowers acquired German rocketry experts and hardware. Early successes in space were viewed as a measure of a nation's scientific, economic, and military strength.

Key events and achievements

The Soviet Union achieved several early milestones. These included launching the first artificial satellite, Sputnik 1, in 1957. The Soviets also put the first human, Yuri Gagarin, in orbit in 1961. In response to Sputnik, the U.S. launched Explorer 1 in 1958. The Soviets also achieved the first spacewalk in 1965.

The Apollo program

NASA's Apollo program was designed to achieve the lunar landing. The Apollo spacecraft consisted of the Command Module, Service Module, and Lunar Module. The program used the Saturn V rocket. The Gemini program provided experience for rendezvous and spacewalks.

In Apollo 11 (1969) NASA achieved the first human landing on the Moon.

Legacy of the Space Race and Apollo

The competition drove innovation and became a foundation for cooperation. The rivalry led to international collaboration, including projects like the Apollo-Soyuz Test Project and the International Space Station which have contributed greatly to the technological development of humankind.

Correct answer – 5 Marks. For any other correct/ sensible answer/ explanation, give marks accordingly

17. What are coronal mass ejections (CMEs) and how do they differ from regular solar wind streams?

CME	Solar Wind
Sudden, explosive ejection of large amounts of plasma and magnetic field	Continuous outflow of charged particles from the Sun
500 – 3000 km/s (highly variable)	300 – 800 km/s (relatively steady)
Short-term event (hours to days)	Persistent flow (always present)
Magnetic field instability or reconnection in the solar corona	Continuous expansion of the Sun’s outer atmosphere due to thermal pressure

Correct answer – 5 Marks. For any other correct/ sensible answer/ explanation, give marks accordingly

18. Describe the main objectives and instruments of the Solar Dynamics Observatory.

1. The Helioseismic and Magnetic Imager (HMI)

The Helioseismic and Magnetic Imager (HMI), studies solar variability and characterizes the Sun's interior and the various components of magnetic activity

2. The Extreme Ultraviolet Variability Experiment (EVE)

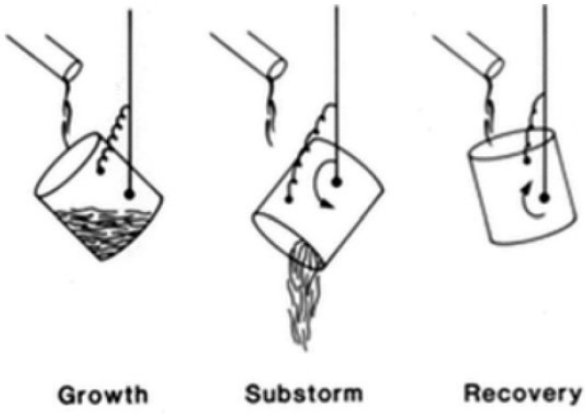
The Extreme Ultraviolet Variability Experiment (EVE) measures the Sun's extreme ultraviolet irradiance with improved spectral resolution

3. Atmospheric Imaging Assembly (AIA)

The Atmospheric Imaging Assembly (AIA) provides continuous full-disk observations of the solar chromosphere and corona in seven extreme ultraviolet (EUV) channels, spanning a temperature range from approximately 20,000 Kelvin to in excess of 20 million Kelvin

Correct answer – 5 Marks. For any other correct/ sensible answer/ explanation, give marks accordingly

19. Explain how the tipping-bucket toy model helps in understanding geomagnetic substorms.



Toy models usually attempt to replicate a system in simple ways. An example of a space weather toy model is the tipping-bucket model of geomagnetic substorms. This model represents the loading and unloading of magnetic energy from the solar wind and IMF into the magnetosphere as water flowing into a bucket that is supported by a spring. Once the amount of energy (water) reaches a critical threshold, the bucket tips over and rapidly dumps the energy(water) from the system (bucket).

Correct answer – 5 Marks. For any other correct/ sensible answer/ explanation, give marks accordingly

20. Explain the processes occurring in the bow shock and magnetosheath.

When the fast-moving solar wind from the Sun reaches a planet with strong magnetic field, it cannot pass straight through the planet's magnetic barrier. Instead, the solar wind suddenly slows down, compresses, and heats up at the bow shock. Also, the solar wind changes from supersonic to subsonic speeds, and its energy is partly converted into heat, making the plasma hot, dense, and turbulent. All these processes result in heating, compression, and deflection of the solar wind plasma around the planet's magnetosphere.

Magnetosheath lies just beyond the bow shock. Within the magnetosheath, the plasma continues to flow turbulently around the magnetosphere, exhibiting fluctuations in velocity, density, and magnetic field. IMF lines become draped and compressed around the magnetosphere, and various magnetic reconnection processes may occur near the magnetopause, allowing partial mixing of solar wind particles into the magnetosphere. The magnetosheath thus acts as a buffer zone, dissipating the energy of the solar wind and providing a transition region between the shocked solar wind and the stable magnetic environment of the magnetosphere. Overall, the bow shock and magnetosheath work together as a protective shield, reducing the strength of the solar wind and ensuring that the planet's magnetosphere remains stable and safe from the Sun's continuous stream of charged particles.

Correct answer – 5 Marks. For any other correct/ sensible answer/ explanation, give marks accordingly

21. Explain the variation of different layers of earth's ionosphere during night as compared to day.

Photons of differing energies are able to penetrate and interact with atoms and molecules in Earth's atmosphere. Densities of atmospheric constituents also vary with height and therefore the ionosphere forms a number of different regions at different altitudes above the surface of Earth.

D region

the lowest ionospheric, extends from approximately 50 km to 90 km

The sources of ionization - solar UV photons ionizing nitric oxide, ionization produced by cosmic rays produce

The D region is essentially only present during the day (though cosmic rays produce a residual level of ionization at night) and the level of ionization in the D region is the lowest of the different regions of the ionosphere.

During Solar storms the D region ionization can become so intense that HF radio communication is completely blacked out.

E region

Originally called the Kennelly– Heaviside layer- extends from 90 km to 120 km.

The sources of ionization - low-energy (or soft) X-rays and UV solar radiation ionization of molecular oxygen (O_2)

The peak density in the E region is over 100 times greater than the peak density in the D region because recombination is less prevalent at these high altitudes.

E region decays away at night, which effectively raises its height as the faster recombination times at lower altitudes decay the E region away more quickly at low altitudes than at higher altitudes. Besides solar photons, E region ionization also occurs due to energetic particles precipitating into the atmosphere. Particle precipitation-induced ionization increases the E region ionosphere substantially, particularly at night when photon production is absent.

Sporadic E layer is generally short-lived and spatially narrow regions of high ionization at E region altitudes. Sources range from complex dynamics due to the effects of the motion of the neutral atmosphere to meteors entering the upper atmosphere, which burn up and give enough energy to the surrounding neutral gas to create an ionized trail. Sporadic E can last from a few minutes to several hours. Ionization can be locally very high, and therefore high-frequency radio waves can be reflected off these trails for long-distance communication.

F region

The densest region of the ionosphere

It extends from 120 km and usually peaks at 300 km.

The F region is formed by extreme UV solar radiation ionizing atomic oxygen. F layer ionization decreases at night, but not as much as the E and D layer ionization because at this higher altitude recombination rates are lower and the layer consists of atomic oxygen rather than the molecular ions that dominate in the D and E regions. Atomic ions have much lower recombination rates in general than molecular ions. The F layer divides into two layers during the day because of the enhanced photoionization at high altitudes. The F2 peak is more dense than the F1 peak.

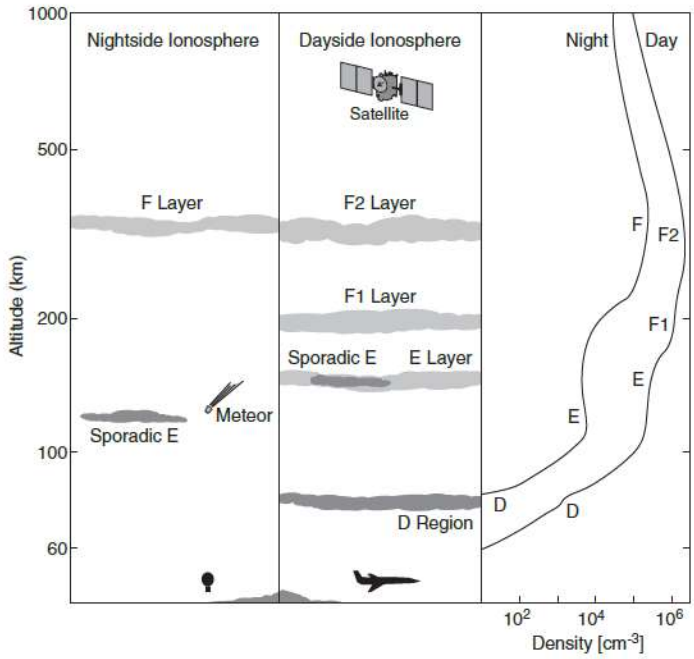


Figure: The vertical structure of Earth's ionosphere.

Correct answer – 5 Marks. For any other correct/ sensible answer/ explanation, give marks accordingly

22. Explain how the solar wind interacts with Earth's magnetic field to create the magnetosphere.

Magnetosphere is created when Earth's magnetic field interacts with solar wind. The solar wind exerts a pressure on the magnetic field compressing it on the day side and stretching it into a long tail.
The interaction of Earth's magnetic field with the magnetized solar wind is similar to that of a rock in a stream. The solar wind (stream) encounters Earth's magnetosphere (rock) as an obstacle and moves around it, leaving a wake behind. In the case of Earth and the solar wind, the interaction produces a long magnetotail. Because the solar wind is supersonic, a shock wave is formed upstream or on the dayside of the magnetosphere. This shock wave is called the bow shock. The bow shock slows the solar wind and begins to divert it around the magnetosphere.

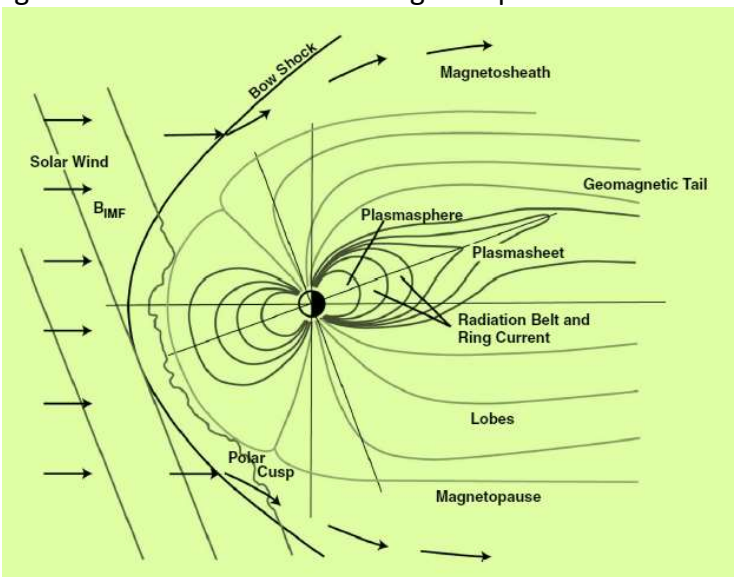


Figure : A noon–midnight cross section of Earth's magnetosphere. Note the dipole shape of the inner magnetosphere. The Sun (hence noon) is to the left and north is up.

Proper explanation with the discussion of bow shock, shock waves, Magnetic reconnection, magnetotail, magnetosheath, magnetopause etc, can be given full mark(figure is not compulsory)

Correct answer – 5 Marks. For any other correct/ sensible answer/ explanation, give marks accordingly

[5x6=30]